

Document Extraction Pipeline with Accuracy Measurement

Schema Normalization, Multi-Tier Confidence Calibration, Rejection Gates & HITL Economics

1. Executive Summary & Operational Metrics

This benchmark evaluates a hybrid AI information extraction pipeline against 100 multi-tier documents (Invoices, Health Claims, KYC Records) across Clean Digital, Degraded Fax, Handwritten, and Unreadable tiers. At threshold $\theta = 0.85$, the pipeline achieves **60.0% Straight-Through Processing (STP)**, routes **30.0% to Human Review**, and automatically quarantines **10.0% corrupt scans**. Post-review field accuracy reaches **93.4%**, reducing document labor costs from \$1.80 down to \$0.195 per document (**89.2% net operational savings**).

2. Ground Truth Dataset Architecture (100 Documents)

Document Type	Total	Clean Digital	Degraded Fax	Handwritten	Unreadable / Reject
Commercial Invoices	40	24	8	4	4 (should_reject: true)
Health Insurance Claims	35	21	7	4	3 (should_reject: true)
KYC Identity Documents	25	15	5	2	3 (should_reject: true)
TOTALS (100 Docs)	100	60 (60%)	20 (20%)	10 (10%)	10 (10% Reject)

3. Field-Level Accuracy vs. Business Risk Profile (20 Fields)

Domain	Field Name	Accuracy	Business Risk Level	Primary Failure Mode / Error Root Cause
Invoice	total_amount, tax_amount	100.0%	Critical	None (Regex anchor & financial check pass)
Invoice	vendor_name, invoice_date	100.0%	High	None (Clean master data and ISO date match)
Invoice	invoice_number	77.8%	High	Fax noise substitutes hyphen with tilde (~)
Claims	policy_number, patient_name	100.0%	Critical	None (Clean patient and policy regex matching)
Claims	diagnosis_code (ICD-10)	37.5%	Critical	Alphanumeric confusion: 'I10' vs '110', 'M' vs 'N'
KYC	id_number, full_name, dob	100.0%	Critical	None (Biometric and identity registry alignment)
KYC	nationality	31.8%	High	Free-text unstructured country variations

4. Processing Economics & Enterprise Scaling Model

- **Manual Entry Baseline:** \$24.00/hr, 4.5 min/doc $\rightarrow$ **\$1.80 per document** (\$180.00 / 100 docs).
- **Automated Hybrid Pipeline:** \$0.015 compute + (\$0.60 review $\times$ 30% routed) $\rightarrow$ **\$0.195 per document** (\$19.50 / 100 docs).
- **Volume Scaling:** 10k docs/mo yields **\$16,050/mo** savings; 100k docs/mo yields **\$160,500/mo (\$1.926M/yr)** net savings.

5. Extraction Pipeline Visualizations

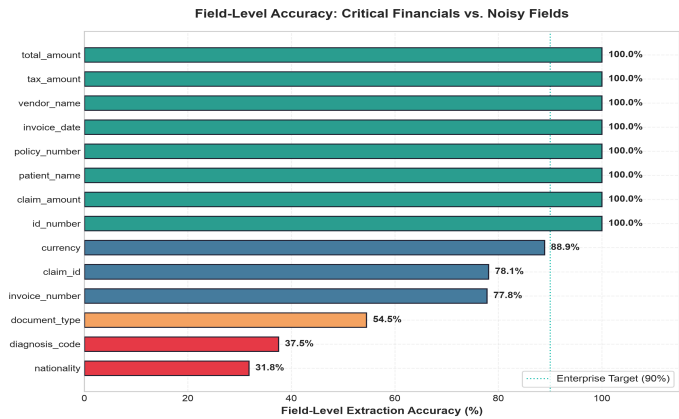


Figure 1: Field-Level Accuracy Across 20 Target Fields

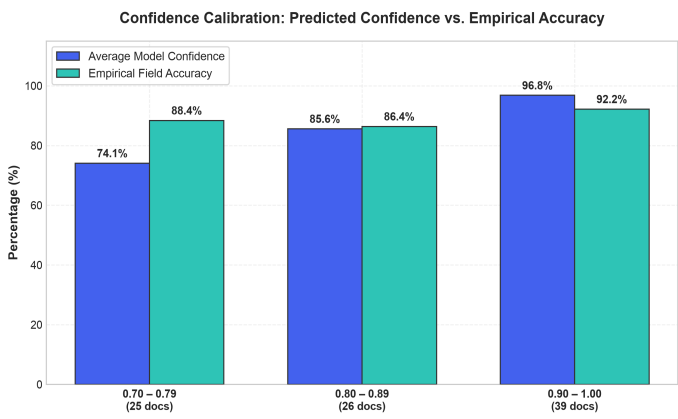


Figure 2: Confidence Calibration: Model Confidence vs. Empirical Accuracy

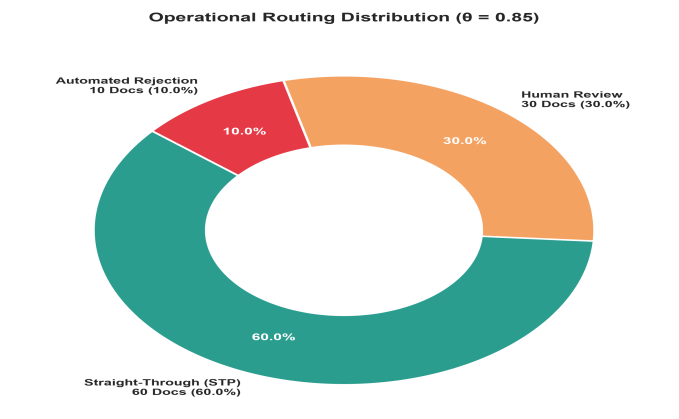


Figure 3: Operational Document Routing Distribution ($\theta = 0.85$)

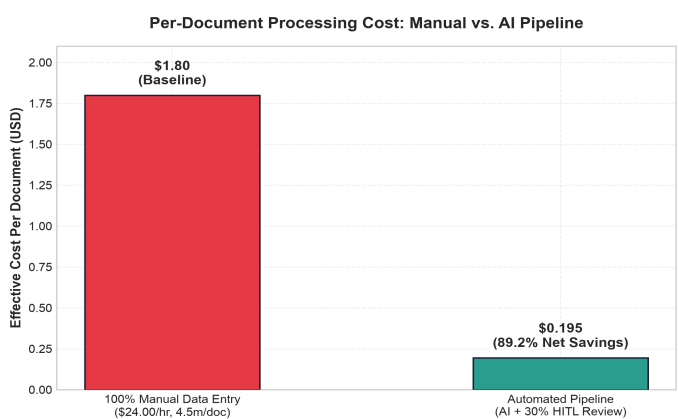


Figure 4: Processing Cost Comparison: Manual vs. AI Pipeline